SIMATS ENGINEERING

LAB EXPERIMENTS

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

COURSE CODE: CSA05

EXP 8 : Query with Joins – EquiJoin, InnerJoin, OuterJoin

```
mysql> create database labexp8;
Query OK, 1 row affected (0.06 sec)

mysql> use labexp8;
Database changed
mysql> create table students (studentid int , studentname varchar(54),deptid int);
Query OK, 0 rows affected (0.07 sec)

mysql> create table department (deptname varchar(54),deptid int);
Query OK, 0 rows affected (0.07 sec)

mysql> insert into student values(1, 'Arun', 101);
ERROR 1146 (42S02): Table 'labexp8.student' doesn't exist
mysql> insert into students values(1, 'Arun', 101);
Query OK, 1 row affected (0.05 sec)

mysql> insert into students values(2, 'Bala', 102);
Query OK, 1 row affected (0.05 sec)

mysql> insert into students values(3, 'Charan', 103);
Query OK, 1 row affected (0.01 sec)

mysql> (4, 'Divya', 101);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds
 to your MySQL server version for the right syntax to use near '4, 'Divya', 101)' at line 1
mysql> insert into students values(4, 'Divya', 101);
Query OK, 1 row affected (0.06 sec)

mysql> select * from students;
+-----------+-------------+--------+
| studentid | studentname | deptid |
+-----------+-------------+--------+
|         1 | Arun        |    101 |
|         2 | Bala        |    102 |
|         3 | Charan      |    103 |
|         4 | Divya       |    101 |
+-----------+-------------+--------+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Department VALUES('CSE',101);
Query OK, 1 row affected (0.06 sec)

mysql> INSERT INTO Department VALUES('ESE',102);
Query OK, 1 row affected (0.04 sec)

mysql> INSERT INTO Department VALUES('IT',103);
Query OK, 1 row affected (0.05 sec)

mysql> select * from department;
+----------+--------+
| deptname | deptid |
+----------+--------+
| CSE      |    101 |
| ESE      |    102 |
| IT       |    103 |
+----------+--------+
3 rows in set (0.00 sec)

mysql> select students , department where students.deptid = department.deptid;
ERROR 1054 (42S22): Unknown column 'students' in 'field list'
mysql> select * from students , department where students.deptid = department.deptid;
+-----------+-------------+--------+----------+--------+
| studentid | studentname | deptid | deptname | deptid |
+-----------+-------------+--------+----------+--------+
|         1 | Arun        |    101 | CSE      |    101 |
|         2 | Bala        |    102 | ESE      |    102 |
|         3 | Charan      |    103 | IT       |    103 |
|         4 | Divya       |    101 | CSE      |    101 |
+-----------+-------------+--------+----------+--------+
4 rows in set (0.00 sec)

mysql> select * from students left join department on students.deptid = department.deptid;
+-----------+-------------+--------+----------+--------+
| studentid | studentname | deptid | deptname | deptid |
+-----------+-------------+--------+----------+--------+
|         1 | Arun        |    101 | CSE      |    101 |
|         2 | Bala        |    102 | ESE      |    102 |
|         3 | Charan      |    103 | IT       |    103 |
|         4 | Divya       |    101 | CSE      |    101 |
+-----------+-------------+--------+----------+--------+
4 rows in set (0.05 sec)
```

```
mysql> select * from students right join department on students.deptid = department.deptid;
+-----------+-------------+--------+----------+--------+
| studentid | studentname | deptid | deptname | deptid |
+-----------+-------------+--------+----------+--------+
|         4 | Divya       |    101 | CSE      |    101 |
|         1 | Arun        |    101 | CSE      |    101 |
|         2 | Bala        |    102 | ESE      |    102 |
|         3 | Charan      |    103 | IT       |    103 |
+-----------+-------------+--------+----------+--------+
4 rows in set (0.00 sec)

mysql> select * from students cross join department;
+-----------+-------------+--------+----------+--------+
| studentid | studentname | deptid | deptname | deptid |
+-----------+-------------+--------+----------+--------+
|         1 | Arun        |    101 | IT       |    103 |
|         1 | Arun        |    101 | ESE      |    102 |
|         1 | Arun        |    101 | CSE      |    101 |
|         2 | Bala        |    102 | IT       |    103 |
|         2 | Bala        |    102 | ESE      |    102 |
|         2 | Bala        |    102 | CSE      |    101 |
|         3 | Charan      |    103 | IT       |    103 |
|         3 | Charan      |    103 | ESE      |    102 |
|         3 | Charan      |    103 | CSE      |    101 |
|         4 | Divya       |    101 | IT       |    103 |
|         4 | Divya       |    101 | ESE      |    102 |
|         4 | Divya       |    101 | CSE      |    101 |
+-----------+-------------+--------+----------+--------+
12 rows in set (0.05 sec)
```